

Topic: Towards Traceable Stablecoins in Decentralized Finance

1. Introduction

The success of cryptocurrencies such as Bitcoin and Ethereum has made blockchain technology a prominent research topic in distributed systems, motivating the adoption of distributed ledger infrastructures for mission-critical applications. One prominent outcome of this development is the emergence of Decentralized Finance (DeFi), which reimagines financial services on blockchain platforms. While DeFi enhances efficiency and reduces reliance on trusted intermediaries through decentralization, it simultaneously challenges conventional regulatory and compliance frameworks that depend on centralized oversight and identity-based supervision. Stablecoins are blockchain-based digital assets whose value is designed to remain stable by being pegged to external references. They have become a cornerstone of the DeFi ecosystem, serving as the primary medium of exchange, store of value, and settlement asset across decentralized finance applications. Their relative price stability makes them essential for mitigating volatility in on-chain markets and for bridging traditional financial systems with decentralized platforms.

However, most stablecoins used in DeFi lack effective transaction traceability. Transactions are typically conducted between participants identified only by pseudonymous blockchain addresses, with no reliable linkage to real-world identities. As a result, the absence of built-in traceability mechanisms hinders regulators and compliance entities from effectively tracking fund flows or enforcing consistent oversight across decentralized protocols. This lack of traceability poses significant challenges to Anti-Money Laundering (AML) and Counter-Terrorist Financing (CFT) efforts. Stablecoins can be exploited to launder illicit funds, evade sanctions, and finance criminal or terrorist activities through rapid, cross-border transfers and complex on-chain transaction patterns. Such misuse not only threatens financial integrity and public security but also undermines confidence in DeFi systems, increasing regulatory pressure that could impede their long-term development.

2. Objectives

In the face of these challenges, this work aims to explore mechanisms for enhancing the traceability of stablecoin transactions in decentralized finance. Specifically, our objectives are summarized as follows:

1. **Characterize the traceability gap in DeFi stablecoins.** We systematically examine existing stablecoin designs and transaction flows to uncover structural factors that impede effective monitoring and regulatory oversight.
2. **Design a framework for traceable stablecoins.** We propose mechanisms that enable transaction tracking while preserving the decentralized nature of blockchain participants.
3. **Evaluate the security and practicality of traceable solutions.** We evaluate the proposed mechanisms for their effectiveness against AML/CFT risks and their integration with current DeFi protocols.